

Agent t Plan (Client Ready)

Budget target: \$50-\$100/month | Channels included: Web + Wha

1) Positioning for Clients

Project Title: AI Customer Support Agent for ERP-enabled SaaS (Automates 60-75% repetitive support queries).

Problem: ERP support teams lose time on repetitive questions (order status, invoices, account updates). This increases response time and support cost.

Solution: A business-ready AI support system using low-cost LLM inference, retrieval from your knowledge base, and secure ERP API integration with human handoff.

2) Cheapest Practical Architecture

Layer	Lean Choice
UI	Website chat + WhatsApp + Slack (all available)
Backend	.NET API on your existing i7 server
Retrieval	PostgreSQL + pgvector on same server
LLM	Single low-cost external API model
Memory	Session table in Postgres (no Redis initially)
Handoff	Create ticket + notify human agent

This setup avoids cloud GPU costs while keeping response quality acceptable for ERP support.

3) Exact Lean Cost Plan (5,000 Conversations/Month)

Category	Line Item	Monthly Cost (USD)
LLM	Low-cost model usage (tight token limits)	\$38
Database	Postgres + pgvector (self-hosted)	\$0
Storage/Backups	Nightly backup + object storage	\$8
Monitoring	Basic uptime + error logging	\$4
Channels	WhatsApp provider + low-volume messaging	\$28
Channels	Slack bot/webhook operations	\$5
Infra Overhead	Networking/misc ops reserve	\$9
TOTAL	Exact monthly total (all 3 channels active)	\$92

Cost per conversation: $\$92 / 5,000 = \0.018

Assumption: average ~1,500 input tokens and ~250 output tokens with strict prompt control, top-3 retrieval, and response cap.

4) Cost Control Rules (Required to Stay Below \$100 with 3 Channels)

- Limit max output tokens per answer (e.g., 120-180).
- Use retrieval top-k = 3 (avoid long context packing).
- Use one low-cost model only in Phase 1.
- Add answer cache for repetitive FAQs.
- If confidence is low, handoff to ticket instead of multi-turn loops.
- Set monthly hard cap with alert at 70% usage.
- Enforce WhatsApp template minimization and short response policy.

5) Implementation Plan (4 Weeks)

Week	Deliverables	Success Check
Week 1	Intent mapping + ERP API wrappers + KB cleanup	Top 20 intents mapped
Week 2	RAG ingestion + prompt template + web chat widget	Accurate answers on test set
Week 3	Handoff flow + WhatsApp + Slack integration + logging	All channels pass UAT
Week 4	UAT, tuning, go-live	Cost/day within target range

6) Upgrade Path (After Traction)

Phase 2: Add premium fallback model only for complex cases once ROI is validated.

Phase 3: Add richer channel automation (broadcasts, advanced workflows) as budget grows.

Final Recommendation: Run all three channels from day one with strict token and messaging limits. This keeps estimated monthly spend near \$92 while maintaining decent support performance.